

# Reason-WAM: Vision-Language Reasoning for World Action Models in Urban Aerial Goal Navigation

Anonymous ACL submission

## Abstract

Urban aerial goal navigation requires an embodied agent to infer a long-horizon route from sparse natural-language goals while executing geometrically precise actions in a large-scale 3D environment. Existing large multi-modal models exhibit strong semantic reasoning and high-level planning, but they often fail to convert spatial intent into stable low-level control. Conversely, recent World Action Models (WAMs) based on video-generation world models can predict short-horizon visual futures and decode them into executable actions, but remain limited in multi-step logical planning. We propose **Reason-WAM**, a hierarchical vision-language navigation framework that couples a reasoning-oriented vision-language model (VLM) planner with a world-action executor. The VLM observes a six-view panoramic context and decomposes a goal into subgoals or exploratory primitive commands, while the WAM predicts short-horizon embodied observation sequences and decodes them into actions for each subgoal. We further introduce a bottom-up training recipe: first train visual odometry and the world-action module from human flight trajectories, relabel trajectory segments into route instructions using a VLM, and then optimize the VLM planner with a multi-turn agentic GRPO objective using task completion and distance-to-goal progress as rewards. Reason-WAM bridges high-level visual-language reasoning and low-level spatial action, providing a scalable route toward long-horizon object navigation in urban airspace.

## 1 Introduction

A central goal of embodied intelligence is to enable robots to act upon natural-language intentions: a user specifies what should be achieved, and the agent autonomously determines how to achieve it through perception and action. Goal-oriented navigation is a canonical instance of this

language-to-action capability. Given a natural-language goal such as “a nearby convenience store,” the agent is not provided with a step-by-step route; instead, it must continuously observe the environment, infer where the target is likely to appear, and move toward it until success. This setting becomes particularly challenging in urban aerial environments, where the agent must operate in large-scale 3D space under partial observation. We observe that successful goal-oriented aerial navigation requires two complementary capabilities:

- **Spatial reasoning.** Since no route is specified, the agent must infer a goal-directed exploration strategy from visual observations and task context. Such reasoning should guide the agent toward likely target regions, actively reveal informative visual cues, and update the plan as new observations become available. For example, to find a convenience store, an agent may first move toward a road to expose storefronts, inspect shop signs, and then approach the most likely target.
- **Spatial action grounding.** Once an intermediate navigation intent is determined, the agent must realize it as executable actions grounded in the surrounding 3D environment. This requires accounting for viewpoint changes, metric distance, altitude, orientation, local geometry, and the spatial relations between the agent and visible landmarks. For instance, a subgoal such as “move toward the road” must be translated into an action sequence that follows the current visual layout and moves the agent toward the intended region.

Large-scale pretrained models have opened new opportunities for embodied spatial intelligence, yet existing paradigms only partially address these two capabilities. LLM- or VLM-based agents

use foundation models as high-level decision makers, enabling semantic interpretation, common-sense reasoning, and flexible planning over navigation goals. However, these agents typically rely on handcrafted prompts or task-specific mapping tools, which limits their generality. Vision-language-action (VLA) models offer a more direct mapping from observations and language to actions. By operating over semantic tokens, they inherit strong abstract reasoning ability from pre-trained vision-language models; nevertheless, deriving precise action tokens from highly compressed semantic representations remains difficult. This compression weakens action grounding in continuous 3D space, where navigation requires fine-grained sensitivity to viewpoint, distance, orientation, altitude, and local scene geometry. More recently, world models and World Action Models (WAMs) provide a complementary direction by predicting short-horizon visual futures from the current observation and a local instruction, and then inversely decoding the predicted transition into actions. This prediction-and-decoding process makes WAMs well suited for grounding concrete short commands in the surrounding 3D environment. Yet, WAMs lack high-level task reasoning: they require an appropriate local instruction as input and are not designed to decompose an open-ended navigation goal into a sequence of executable subgoals.

To this end, we propose **Reason-WAM**, a hierarchical architecture that couples VLM-based spatial reasoning with selective WAM-based action grounding. Given a navigation goal, the VLM reasons over current observations, interaction history, and the goal description to iteratively plan the next step. Each step can take one of three forms: acquiring additional surrounding observations at the current location, issuing a primitive control action, or producing a short route-oriented VLN instruction. Reason-WAM directly executes the first two lightweight forms for efficiency, while delegating VLN-instruction steps to the WAM for grounded execution. The world model predicts how the embodied visual state would change if the instruction were executed, and the predicted transition is converted by a action-decoding module into an executable action sequence. To instantiate this architecture, we design a bottom-up training pipeline: we first train the action-decoding module from video-trajectory pairs, then train the world model with short-horizon route-

oriented instruction-video pairs, and finally optimize the VLM with multi-turn GRPO to learn goal decomposition and selective WAM invocation.

Our main contributions are as follows:

- We propose **Reason-WAM**, a hierarchical architecture that couples VLM-based spatial reasoning with WAM-based action grounding for goal-oriented urban aerial navigation.
- We develop a bottom-up training framework that learns visual odometry, short-horizon world-action dynamics, and VLM-based goal reasoning with multi-turn GRPO.
- We show that Reason-WAM improves open-ended aerial goal navigation with stronger spatial grounding, more efficient execution, and better task decomposition.

## 2 Related Work

**Vision-Language Navigation (VLN).** Vision-language navigation studies closed-loop perception and action under natural-language guidance. Existing settings can be broadly divided into route-oriented navigation, where instructions provide explicit step-by-step movement cues, and goal-oriented navigation, where only a target description is given. The latter is more challenging, requiring the agent to use spatial commonsense reasoning to derive feasible plans and translate the action intent of each step into executable actions.

**Vision Language Models for VLN.** Large language models and vision-language modmels have been increasingly used in embodied navigation, either as agentic planners that decompose goals and maintain semantic memory, or as multimodal backbones in vision-language-action policies that directly predict actions. Their strength lies in semantic abstraction and instruction-level reasoning, enabling high-level decisions such as where relevant objects are likely to appear. However, such models still struggle with fine-grained spatial action and continuous control, often exhibiting unstable orientation, weak metric awareness, and unreliable conversion from semantic intent to executable 3D actions.

**World Action Models for VLN.** World Action Models (WAMs) offer a predictive action-grounding paradigm that first emerged in manipulation, where action-conditioned future states support planning and inverse dynamics. This idea has

recently been extended to navigation by predicting visual futures under candidate instructions or actions and decoding executable movements from the predicted consequences. Such models provide strong local spatial priors and improve short-horizon action grounding, but remain less suited to open-ended goal navigation: their visual predictive basis lacks abstract chain-of-thought reasoning, while prevailing diffusion-based architectures struggle with long-horizon generation.

### 3 Method

For goal-oriented navigation, each episode provides a natural-language goal  $g$ , an initial embodied observation  $o_0$ . At step  $t$ , the agent observes visual input  $o_t$ , maintains an interaction history  $h_t$ , and executes an action  $a_t$  that updates its hidden pose by  $p_t = F(p_{t-1}, a_t)$ . An episode is successful if the final pose reaches the target region  $p^*$ , i.e.,  $\|p_T - p^*\|_2 \leq \epsilon$ . To solve this problem, Reason-WAM combines a hierarchical model architecture with a bottom-up training framework: a VLM performs goal reasoning and step selection, while a WAM grounds route-oriented steps into executable aerial actions.

#### 3.1 Model Architecture

Reason-WAM factorizes navigation into VLM-based spatial reasoning and WAM-based action grounding. At each decision point, the VLM  $\pi_\theta$  conditions on the goal  $g$ , the current embodied observation  $o_t$ , and the interaction history  $h_t$  to reason about an overall plan  $P_t$  and instantiate its next executable step  $s_t$ :

$$(P_t, s_t) \sim \pi_\theta(\cdot \mid g, o_t, h_t), \quad (1)$$

$$s_t \in \mathcal{S}_{\text{sense}} \cup \mathcal{S}_{\text{ctrl}} \cup \mathcal{S}_{\text{route}}.$$

Here  $P_t$  captures the VLM’s current high-level reasoning over the goal and scene, while  $s_t$  is the step to be executed at the current turn. The step space is organized by execution interface:  $\mathcal{S}_{\text{sense}}$  requests additional surrounding observations when the current view is insufficient;  $\mathcal{S}_{\text{ctrl}}$  contains primitive controls for immediate local adjustment; and  $\mathcal{S}_{\text{route}}$  contains short route-oriented VLN instructions whose spatial intent must be grounded over multiple steps. Although the first two types could also be mediated by a world model, they correspond to relatively fixed sensing or primitive-control operations and are therefore executed directly for efficiency, whereas route instructions are delegated to the WAM for action grounding.

For a VLN-instruction step  $s_t \in \mathcal{S}_{\text{route}}$ , the step generally corresponds to a route-oriented navigation instruction, denoted as  $\ell_t$ . The world model  $\pi_\phi$  serves as an intermediate predictive interface: instead of committing to low-level controls immediately, it first forecasts the visual consequences of following the instruction. This converts the abstract route intent into an explicit future-state target, indicating how the embodied observation should change if the instruction is correctly executed. The predicted visual rollout provides dense spatial cues for action grounding, which bridging route semantics and low-level motion. The process can be formalized as:

$$\hat{o}_{t+1:t+H} \sim \pi_\phi(\cdot \mid o_t, s_t), \quad (2)$$

where  $H$  is the execution horizon, and  $\hat{o}_{t+1:t+H}$  denotes a sampled future observation sequence. The resulting visual transition is decoded into executable aerial actions by a learning-based visual odometry model, which estimates relative motion from  $o_t$  and the predicted rollout. Its learned robustness to appearance changes and prediction artifacts makes it suitable for using world-model outputs rather than real future observations:

$$a_{t:t+H-1} = D_\psi(o_t, \hat{o}_{t+1:t+H}). \quad (3)$$

After execution, the new observation is derived, enabling the VLM to re-plan in a closed loop.

#### 3.2 Bottom-UP Training Framework

We adopt a bottom-up training curriculum that reflects the dependency structure of embodied navigation. Each upper-level module relies on a reliable lower-level abstraction: the VLM manager depends on the world model to translate semantic intent into visual predictions, while the world model depends on the action decoder to connect visual transitions with executable motions. This staged training design decouples metric action grounding from semantic decision making, reduces long-horizon credit assignment, and makes the final policy easier to optimize.

Concretely, suppose that each object navigation case in a training dataset contains a navigation goal  $g$ , a human flight video  $o_{0:T}^*$ , and the corresponding action-coordinate sequence  $a_{0:T-1}^*$ . The training process is as follows:

**Action-decoding training.** For a trajectory of length  $T$ , we uniformly partition it into segments

of length at most  $H$ ; trajectories shorter than  $H$  are kept intact. This yields video–action pairs for training the action decoder. Given a trajectory segment  $(o_{i:j}^*, a_{i:j-1}^*)$ , we train the decoder  $D_\psi$  as visual odometry or inverse dynamics:

$$\mathcal{L}_{\text{dec}}(\psi) = \sum_{(i,j)} \|D_\psi(o_i^*, o_{i+1:j}^*) - a_{i:j-1}^*\|_1. \quad (4)$$

This stage trains  $D_\psi$  to map embodied visual transitions into executable UAV motions.

**World-model training.** Each segmented video is relabeled by a VLM into a concise route-oriented step  $s_{i:j}^* \in \mathcal{S}_{\text{route}}$  describing the local movement intent. The resulting step–video pairs supervise the world model  $\pi_\phi$  to generate the future visual sequence from the initial observation and route step:

$$\mathcal{L}_{\text{wm}}(\phi) = - \sum_{(i,j)} \log \pi_\phi(o_{i+1:j}^* | o_i^*, s_{i:j}^*). \quad (5)$$

This objective teaches  $\pi_\phi$  short-horizon instruction-conditioned visual dynamics.

**VLM training with multi-turn GRPO.** After  $D_\psi$  and  $\pi_\phi$  are trained, we optimize the VLM manager  $\pi_\theta$  in multi-turn rollouts. At each decision state, the VLM samples  $(P_t, s_t)$  conditioned on  $(g, o_t, h_t)$ ; the selected step  $s_t$  is executed either directly or through the WAM according to its type. The reward combines final task success with step-wise progress toward the target:

$$R(\tau) = \lambda_s \mathbb{I}[\|p_T - p^*\|_2 \leq \epsilon] + \lambda_p \sum_t (d_{t-1} - d_t), \quad (6)$$

For each decision state  $(g, o_t, h_t)$ , we sample a group of VLM outputs  $(P_{k,t}, s_{k,t})$  and compute group-normalized advantages  $\hat{A}_{k,t}$  from their rollout rewards. The VLM is then updated with a clipped GRPO objective:

$$\mathcal{L}_{\text{grpo}}(\theta) = -\mathbb{E}_{k,t} \left[ \min \left( \rho_{k,t} \hat{A}_{k,t}, \text{clip}(\rho_{k,t}, 1 - \eta, 1 + \eta) \hat{A}_{k,t} \right) \right], \quad (7)$$

where  $\rho_{k,t} = \pi_\theta(P_{k,t}, s_{k,t} | g, o_t, h_t) / \pi_{\text{old}}(P_{k,t}, s_{k,t} | g, o_t, h_t)$ . This bottom-up procedure first trains  $D_\psi$  to ground visual transitions into actions, then learns instruction-conditioned visual dynamics with  $\pi_\phi$ , and finally trains  $\pi_\theta$  to decompose open-ended goals and selectively invoke WAM execution.

## 4 Experiments

### 4.1 Experimental Setup

We plan to evaluate Reason-WAM on the urban aerial goal-oriented navigation benchmark introduced by Zhao et al. (2026). The benchmark contains 5,037 human-collected trajectories in a realistic urban simulator and reports Success Rate (SR), Success weighted by Path Length (SPL), and Distance to Goal (DTG). We compare against four categories of baselines: direct action-as-language VLMs, agent-based planners, VLA models, and WAM-only executors such as WorldVLN-style local navigation.

### 4.2 Quantitative Results

This section will report the main SR, SPL, and DTG results across short, medium, and long trajectories. The central comparison should test whether Reason-WAM improves over both VLM-only planning and WAM-only local execution, especially on long-horizon cases that require multi-step semantic reasoning.

### 4.3 Case Analysis

We will analyze representative success and failure cases. Success cases should highlight how the VLM planner decomposes goals into meaningful subgoals, such as moving toward a road before inspecting storefronts, while the WAM executor realizes each subgoal as stable spatial motion. Failure cases should diagnose wrong semantic subgoals, ambiguous visual evidence, accumulated WAM execution errors, or premature stopping.

### 4.4 Ablation Study

The ablations should isolate the contribution of each design:

- **Hierarchy:** VLM-only, WAM-only, and Reason-WAM.
- **Perception context:** single front view versus six-view context.
- **Decision interface:** semantic subgoals only, primitive actions only, and mixed decisions.
- **Training stages:** without route relabeling, without visual-odometry pretraining, and without agentic GRPO.
- **Reward design:** final success only versus success plus distance-to-goal progress.

## 5 Conclusion

We presented Reason-WAM, a hierarchical framework for urban aerial goal navigation that combines VLM-based reasoning with world-action execution. By assigning high-level planning to a six-view VLM manager and short-horizon spatial control to a WAM executor, the method addresses the complementary weaknesses of direct VLM action generation and WAM-only local navigation. The proposed bottom-up training pipeline further connects human demonstrations, route-instruction relabeling, world-model learning, and agentic GRPO planner optimization. This provides a concrete path toward scalable long-horizon object navigation in urban airspace.

## Limitations

Reason-WAM depends on the quality of both VLM planning and WAM execution. If the VLM produces an incorrect semantic subgoal, the WAM may faithfully execute a locally plausible but globally wrong trajectory. Conversely, if WAM predictions are visually plausible but action-inconsistent, low-level errors may accumulate. Real-world UAV deployment also requires safety constraints, geofencing, reliable state estimation, and human supervision. This paper therefore focuses on simulator-based evaluation and treats real-world deployment as future work.

## References

Baining Zhao, Ziyu Wang, Jianjie Fang, Zile Zhou, Yanggang Xu, Yatai Ji, Jiacheng Xu, Qian Zhang, Weichen Zhang, Chen Gao, and Xinlei Chen. 2026. How far are large multimodal models from human-level spatial action? a benchmark for goal-oriented embodied navigation in urban airspace. Manuscript.

## A Planned Prompt Interface

The VLM planner prompt should include: the global goal instruction, six labeled views, a compact memory of previous decisions, whether the last execution improved distance to goal, and a constrained output schema containing subgoal, primitive, or stop. This schema will make multi-turn GRPO credit assignment easier and reduce invalid action outputs.